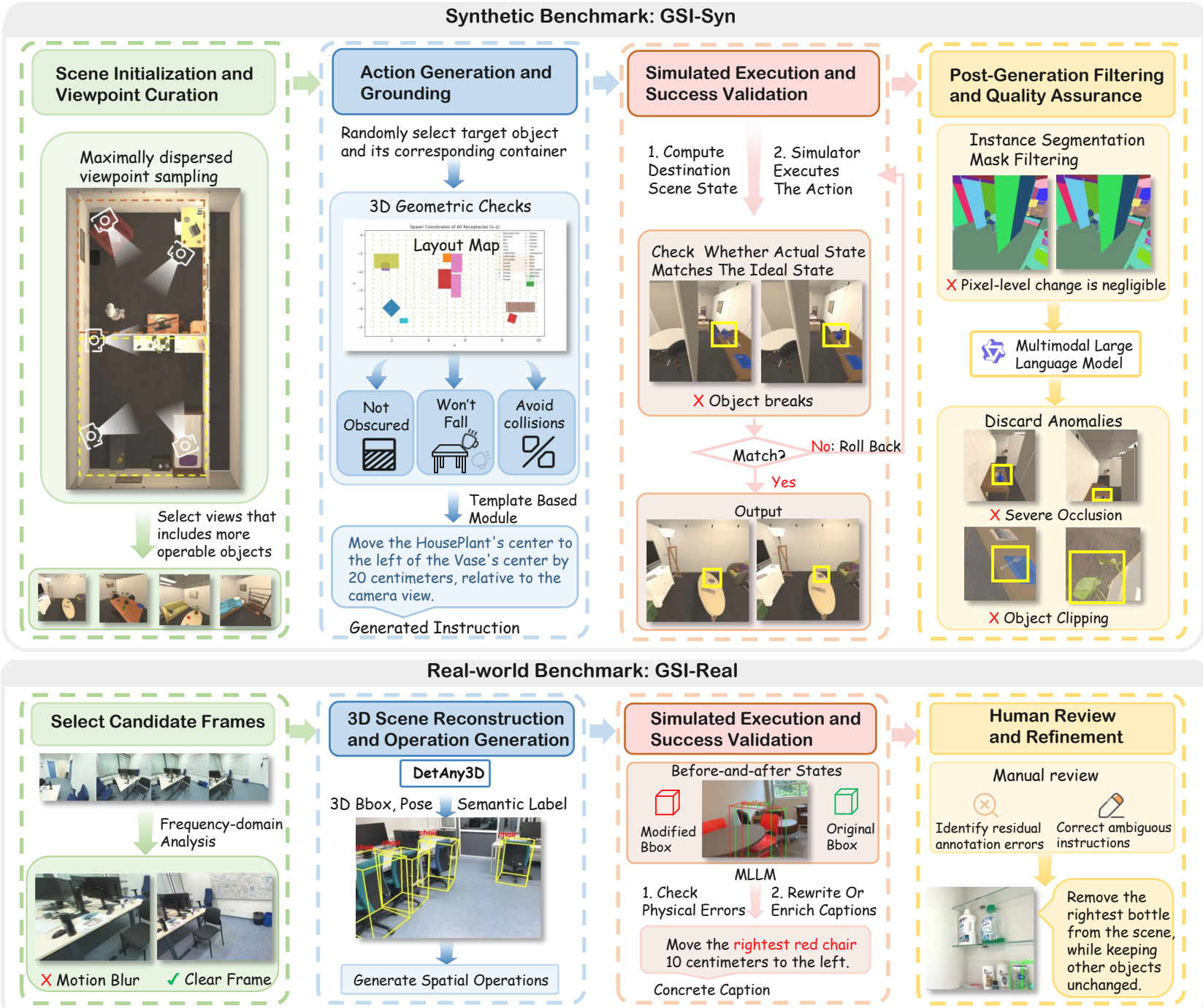
Synthetic Benchmark: GSI-Syn
Scene Initialization and Viewpoint Curation
Maximally dispersed viewpoint sampling
Select views that includes more operable objects
Action Generation and Grounding
Randomly select target object and its corresponding container
3D Geometric Checks
Layout Map
Not Obscured
Won't Fall
Avoid collisions
Template Based Module
Move the HousePlant's center to the left of the Vase's center by 20 centimeters, relative to the camera view.
Generated Instruction
Simulated Execution and Success Validation
1. Compute Destination Scene State
2. Simulator Executes The Action
Check Whether Actual State Matches The Ideal State
X Object breaks
Match?
No: Roll Back
Yes
Output
Post-Generation Filtering and Quality Assurance
Instance Segmentation Mask Filtering
X Pixel-level change is negligible
Multimodal Large Language Model
Discard Anomalies
X Severe Occlusion
X Object Clipping
Real-world Benchmark: GSI-Real
Select Candidate Frames
Frequency-domain Analysis
X Motion Blur
✓ Clear Frame
3D Scene Reconstruction and Operation Generation
DetAny3D
3D Bbox, Pose
Semantic Label
Generate Spatial Operations
Simulated Execution and Success Validation
Before-and-after States
Modified Bbox
Original Bbox
MLLM
1. Check Physical Errors
2. Rewrite Or Enrich Captions
Move the rightest red chair 10 centimeters to the left.
Concrete Caption
Human Review and Refinement
Manual review
Identify residual annotation errors
Correct ambiguous instructions
Remove the rightest bottle from the scene, while keeping other objects unchanged.